

PPH + NRP Combined Case



Section I: Scenario Demographics

Scenario Title:	Post-Partum Hemorrhage + NRP		
Date of Development:	26/02/2016 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 – 2)	☒ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Donika Orlich (NRP from NRP case by Kyla Caners)
Affiliations/Institution(s):	McMaster University
Contact E-mail (optional):	Donika.orlich@medportal.ca

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to severe post-partum hemorrhage and a neonate requiring resuscitation
CRM Objectives:	1) Demonstrates leadership by staying calm during crisis, making prompt decisions and maintaining a global perspective 2) Listens to team input and updates patient with effective communication 3) Demonstrates resource utilization by calling for help appropriately, prioritizing tasks and triaging mother and neonate as required
Medical Objectives:	1) Demonstrates appropriate ED preparation based on EMS patch call for likely precipitous delivery 2) Investigates for the major causes of PPH and demonstrates an approach to addressing each cause 3) Appropriately manages the flat neonate using the NRP guidelines for O2 use, PPV, intubation, chest compressions and medications 4) Describes the procedural steps of umbilical vein catheterization

Case Summary: Brief Summary of Case Progression and Major Events

The team receives advanced notification from EMS about a woman who is imminently delivering. Upon arrival, delivery will be uncomplicated, but the neonate will appear lifeless. Neonatal resuscitation should be initiated. Eight minutes into the neonatal resuscitation, the team leader will be notified that the mother continues to hemorrhage and is becoming hypotensive. They must begin concurrent workup and management of the mother while continuing to run the neonatal resuscitation. Second & third line medical therapies for uterine atony will be needed, and also manual uterine exploration and packing. Early consultation should be made to NICU, ICU, OB, and Interventional Radiology.

References

- Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
- EMedicine: website accessed Feb 26, 2016; <http://emedicine.medscape.com/article/275038-treatment#d11>
- Life in the Fast Lane: website accessed Feb 26, 2016; <http://lifeinthefastlane.com/cc/haemorrhage-and-pregnancy/>
- Sim One Case: Maternal Fetal Hemorrhage downloaded Feb 26, 2016 at <http://www.sim-one.ca/scenario/code-pink-scenario-acute-fetal-maternal-hemorrhage>



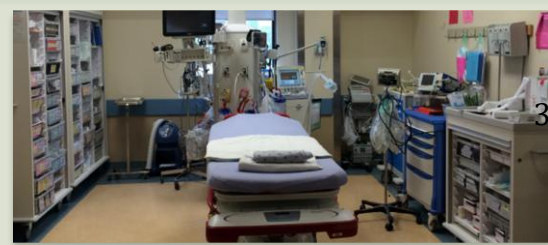
PPH + NRP Combined Case



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☐ Computerized Mannequin	Realism:	☒ Conceptual
	☒ Neonatal Computerized Mannequin		☒ Physical
	☒ Standardized Patient PLUS OB task trainer (hybrid)		☒ Emotional/Experiential
	☐ Other		☐ N/A
Con-federates	Brief Description of Role		
SP	Active labor. Heavy breathing + pain during contractions. Asks about infant upon delivery. Later decreased LOC as bleeding increases.		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Adult + Neonatal Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☒ Blood Products	☒ Adult + Neonatal ET Tubes	☒ Other: Umbilical vein catheter	
☒ Interosseous Set-up	☐ LMA	☒ Other: Blankets, scissors, umbilical clamps	
D. Moulage			
Appropriate moulage as required to have standardized patient set up with OB task trainer between legs. Blood to simulate post-partum hemorrhage (can use a saline bag died red and attached to an iv line in the pelvis – open up the line when the hemorrhage is to start)			
E. Approximate Timing			
Set-Up:	10 min	Scenario:	20 min
		Debriefing:	40 min

PPH + NRP Combined Case



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case			
EMS Patch: "We have a 26 year-old female who is 38 weeks pregnant and appears to be in active labor. She is complaining of severe abdominal pain and has had some vaginal bleeding. We don't see any crowing yet, but the patient feels the baby's head is about to come out. Patient's Vitals as follows: HR 120, BP 140/85, RR 20, O2 100% on room air. ETA 2 minutes."			
B. Patient Profile and History			
Patient Name: Mary Park		Age: 26	Weight: 90kg
Gender: ☐ M ☒ F		Code Status: Full	
Chief Complaint: Active Labor + Vaginal Bleeding			
History of Presenting Illness: Started with cramping about 1 hour ago. Noticed some vaginal bleeding so she called EMS as her husband wasn't home from work yet. Pain is getting worse, and contractions are currently about 1 minute apart.			
Past Medical History:	Healthy G2P1	Medications:	None
	GBS -ve		
Allergies: None			
Social History: Lives with husband and 2 yo daughter. No EtOH, smoking, drugs.			
Family History: No bleeding disorders.			
Review of Systems:	CNS:	Visible pain/discomfort.	
	HEENT:	Nil	
	CVS:	Nil	
	RESP:	Nil	
	GI:	Nil	
	GU:	Severe abdo cramping q1min. Small amount vag bleeding noted. Membranes ruptured with EMS. Feels like the baby's head is "right there".	
	MSK:	Nil	INT: Nil
C. Baseline Simulator State and Physical Exam			
☐ No Monitor Display		☒ Monitor On, no data displayed	
☐ Monitor on Standard Display			
HR: 122/min	BP: 138/80	RR: 22/min	O ₂ SAT: 100%
Rhythm: Sinus Tach	T: 36.5°C	Glucose: 6 mmol/L	GCS: 15 (E4 V5 M6)
General Status: Visible pain/discomfort. Heavy breathing.			
CNS:	Alert and Oriented		
HEENT:	Normal.		
CVS:	Normal heart sounds. No edema.		
RESP:	GAEB. No crackles/wheeze.		
ABDO:	FHR: 150 bmp		
GU:	Sterile Cervical Assessment: head crowning		

PPH + NRP Combined Case



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers- PATIENT #1 (Mother)			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. EMS Patch	Case begins by reading clinical vignette to team as an EMS patch.	<u>Learner Actions</u> - ☐ Prepare for delivery (scissors, clamps, oxytocin, foley, analgesic, suture/needle driver, PPE) - ☐ Prepare for NRP (warmer, blankets, O2, broselow, ETT/suction, infant BMV) - ☐ Early consult to Obstetrics + NICU/Peds + Anesthesia - ☐ Pull nursing/RT staff to increase available team members + Extra ER physician - ☐ US at bedside	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> <u>Triggers</u> <i>For progression to next state</i> - 2 minutes → 2. Patient Arrival
2. Patient Arrival Rhythm: Sinus tach HR: 122/min BP: 138/80 RR: 22/min O ₂ SAT: 100% FHR: 170/min	++ Discomfort, heavy breathing, crying with contractions. Blood on EMS sheet/legs	<u>Learner Actions</u> - ☐ IV, O2, monitors - ☐ Px: FHR monitor (if available), cervical examination - ☐ US to confirm presenting part if unsure - ☐ Labs (CBC, G&S, INR) - ☐ Focused Hx	<u>Modifiers</u> - No attempt at cervical exam by 2mins → patient screams “the head is coming out!!!” <u>Triggers</u> - 3 mins or all learner actions complete → 3. Delivery
3. Delivery Rhythm: Sinus tach HR: 130/min BP: 140/90 RR: 28/min O ₂ SAT: 100%	++ Screaming in pain	<u>Learner Actions</u> - ☐ Encourage pushing with contractions - ☐ Support perineum - ☐ Control delivery of head - ☐ Check for nuchal cord (x1) - ☐ Gentle downward pressure to deliver anterior shoulder - ☐ Oxytocin 10U IM once ant shoulder delivered - ☐ Upward traction to deliver posterior shoulder - ☐ Clamp and cut cord - ☐ Infant to NRP provider - ☐ Delegation of placental delivery to resident or delegation of NRP leadership	<u>Modifiers</u> <u>Triggers</u> - Infant delivered → 4. Post-delivery Begin PATIENT #2 CASE with “baby isn’t moving or breathing”

PPH + NRP Combined Case



4. Post-Delivery HR: 105 BP: 120/80 RR: 12 O2: 100%	Mother crying and asking about her bay Small amount PV bleeding, but comfortable	<u>Learner Actions</u> - ☐ Gentle traction and SP pressure until placenta delivered - ☐ Inspect vessels and placenta integrity - ☐ Trans-abdominal uterine massage - ☐ Update mother +/- assign team member or social work to being with mother	<u>Modifiers</u> - Mother not updated → begins to whale and cry “what is happening to my baby” and tries to get out of the bed - 8 mins into PATIENT #2 → 5. Hemorrhage
5. Hemorrhage Rhythm: Sinus Tachy HR: 125/min BP: 70/50 RR: 18/min O2SAT: 100%	Mother looks pale ++ blood covering the bed	<u>Learner Actions</u> - ☐ 2 nd IV + Level 1 transfuser - ☐ Assess uterine tone - ☐ Assess for uterine, cervical, or vaginal trauma - ☐ Call for 4 unites O- blood - ☐ Activate massive transfusion protocol - ☐ 2L IV NS until blood arrives - ☐ Uterine Massage - ☐ Oxytocin (10u IM or 5u IV) if not already given - ☐ Bimanual Massage - ☐ Ergotamine 0.5mg IM - ☐ Carboprost 0.25mg IM - ☐ Misoprostol 1mg PR - ☐ TXA 1g IV - ☐ Plan for sedation ± intubation - ☐ Uterine exploration ± packing - ☐ Consider DIC screen - ☐ Obstetrics + ICU consult	<u>Modifiers</u> - 2L NS → HR 118, BP 80/65 - Uterotonics → still bleeding - Bimanual compression → clot expelled, still bleeding - 2 minutes → blood arrives - 4u pRBC → HR 110, BP 90/65 - Attempt uterine exploration without analgesia/sedation → patient screams in pain - Uterine exploration → retained products removed <u>Triggers</u> - Retained products removed → 6. Resolution
6. Resolution HR: 105 BP: 95/50 RR: 16 O2: 100%		<u>Learner Actions</u> - ☐ Update Obstetrics - ☐ Consult ICU	<u>Modifiers</u> <u>Triggers</u> - 20 mins or all actions complete → END CASE

PPH + NRP Combined Case



Scenario States, Modifiers and Triggers- PATIENT #2 (Neonate)			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: sinus HR: 40/min BP: ?/? RR: 20/min O ₂ SAT: 58% T: 36.5°C	Flat. Not crying. Minimal respiratory effort. No meconium	<u>Learner Actions</u> - ☐ Dry & warm - ☐ Position airway - ☐ Consider suctioning (mouth then nose) - ☐ Stimulate - ☐ Check HR → start PPV with 21% O ₂ (rate 40-60/min) - ☐ Initiate SpO ₂ monitoring - ☐ Check cap sugar: 5.2	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - PPV → O ₂ 65%, HR 45 <u>Triggers</u> <i>For progression to next state</i> - 1 min → 2. 60 Second Mark
2. 60 Second Mark HR:45 RR: bagging rate O ₂ : 65% *same as above if no BMV started	Unchanged.	<u>Learner Actions</u> - ☐ Re-check HR at 60 sec - ☐ Start CPR at 90 sec - ☐ Change to 100% O ₂ - ☐ Maintain 3:1 ratio for CPR (120 events/min) - ☐ Check HR/RR q1min - ☐ Epi 0.01mg/kg IV after 1 st HR check if still <60 after CPR - ☐ Prepare for intubation	<u>Modifiers</u> - 2 min → O ₂ SAT 65% - 3 min → O ₂ SAT 70% <u>Triggers</u> - 4 min → 3. HR >60
3. HR > 60 HR → 80 RR → 40 O ₂ SAT → 75% BP → 55/30	Appears less grey.	<u>Learner Actions</u> - ☐ Intubate - ☐ Obtain iv access - ☐ Prepare for UVC placement - ☐ IVF bolus (20ml/kg =50ml) - ☐ Check cap sugar (if not yet) - ☐ Send blood work, including blood cultures	<u>Modifiers</u> - 5 min → O ₂ SAT 80% <u>Triggers</u> - 7 min → 4. HR >100 (set to slowly return to normal)
4. HR >100 O ₂ SAT → 83% HR → 120	Appears pink. Moving limbs now.	<u>Learner Actions</u> - ☐ Call NICU - ☐ Finish tasks from above	<u>Triggers</u> - Called to PATIENT #1 for hypotension → 5. Hypotension - 10 minutes or once intubated → END PATIENT #2 case

PPH + NRP Combined Case



Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results- PATIENT #1

None available

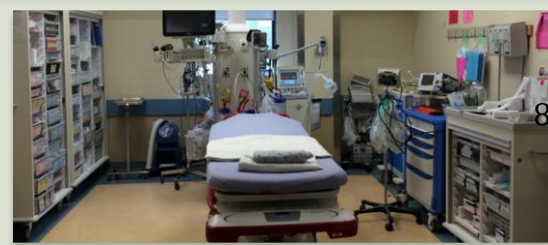
Images (ECGs, CXRs, etc.)

None available

Ultrasound Video Files (if applicable)

None available – describe head down

PPH + NRP Combined Case



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To expose learners to severe post-partum hemorrhage and neonate requiring resuscitation
CRM Objectives:	1) Demonstrates leadership by staying calm during crisis, making prompt decisions and maintaining a global perspective 2) Listens to team input and updates patient with effective communication 3) Demonstrates resource utilization by calling for help appropriately, prioritizing tasks and triaging mother and neonate as required
Medical Objectives:	1) Demonstrates appropriate ED preparation based on EMS patch call for likely precipitous delivery 2) Investigates for the major causes of PPH and demonstrates an approach to addressing each cause 3) Appropriately manages the flat neonate using the NRP guidelines for O2 use, PPV, intubation, chest compressions and medications 4) Describes the procedural steps of umbilical vein catheterization
Sample Questions for Debriefing	
1) What was your approach to managing two simultaneous resuscitations? How did you delegate tasks? Did you use any safety parameters with those you delegated to? 2) What communication strategies did you use to calm the mother? 3) What is your management approach to PPH? What medications are considered 1 st , 2 nd and 3 rd line? What physical maneuvers can be done to investigate and lessen bleeding? 4) What was your goal O2 sat in this neonate? What are the steps to achieve this in neonatal resuscitation?	
Key Moments	
1) Recognizes need to task switch to infant requiring resuscitation and effectively hands over maternal care to junior staff.	
2) Initiates NRP using current guidelines for O2, PPV, intubation, chest compressions and epinephrine use.	
3) Recognizes PPH and initiates blood transfusion, medical therapy and manual maneuvers to investigate and lessen bleeding.	

